

ALVIN EKELUND

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EDUCATION

HARVARD UNIVERSITY

M.S. in Data Science, Harvard Griffin GSAS (Harvard SEAS)

Cambridge, Massachusetts
Aug 2026 - Feb 2028

AALTO UNIVERSITY

B.Sc. in Industrial Engineering and Management; Minor in Computer Science

Espoo, Finland
Aug 2023 - Feb 2026

- GPA 4.9/5.0 (Honors); completed in 2.5 years. Thesis on the productivity impact of LLMs in enterprises and startups, graded 5/5; exchange at National University of Singapore.
- Coursework: Data Structures and Algorithms, Software Engineering, Databases, Web Software Development, Information Security, and Machine Learning.

EXPERIENCE

MIRACLE CONSULTING GROUP

Software Engineer Intern

Helsinki, Finland
May 2026 - Present

- Improve the accuracy and latency of production Graph-RAG pipelines for a Fortune 500 client on Oracle enterprise systems.
- Develop end-user AI and analytics applications and a unified design system across the client's data products.
- Build an internal agent harness that automates workflows across ERP, CRM, HR, and finance systems.

CADENTIA TECHNOLOGIES

Founder's Associate

Helsinki, Finland
Aug 2025 - May 2026

Business Development Intern

Feb 2025 - Aug 2025

- Worked directly with the CEO to research AI knowledge factories, build customer demos, and iterate the product for sales meetings; the demos helped secure the company's first Fortune 500 customer.
- Automated market and competitive research and developed an internal AI video-production pipeline.

APPLE

Research Project

Helsinki, Finland
Jan 2025 - Jun 2025

- Conducted market research on B2B and institutional customer segments for Apple in Finland.

AALTO UNIVERSITY DEPARTMENT OF COMPUTER SCIENCE

Teaching Assistant

Espoo, Finland
Sep 2024 - Jan 2025

- Taught and helped develop an introductory programming course; mentored first-year students in debugging and problem-solving.

SELECTED SOFTWARE PROJECTS

CHAIN-OF-THOUGHT WORKSPACE ABLATION

Aug 2026

- Replicated Anthropic's J-space ablation on Qwen3-4B across GSM8K, MATH-500, and AIME; wrote the ablation, two-stream decoding loop, grading harness, and analysis (PyTorch, Transformers, RunPod A100).
- Pre-registered the experiment and ran about 26 A100-hours; J-space ablation increased MATH-500 error by 8.0 points (95% CI 2.7-13.3), while norm-matched random ablation had no effect.

PERSONAL BRAIN

2026

- Built a local-first knowledge graph and MCP memory server with LLM-powered ingestion, semantic search, graph traversal, Ebbinghaus-style decay, and interactive visualization (Python, SQLite, embeddings).

WHAT PREDICTS AI AUTOMATION

2025 - 2026

- Analyzed 425K Anthropic Economic Index records across 3,259 occupational tasks, finding a 25 percentage-point gap in automation between API and chat deployments (Python, Jupyter, O*NET).

P-PASCAL - NUS HACK & ROLL

Jan 2026

- Built and shipped a full-stack social prediction-market app for friend groups, combining LLM-derived empirical priors with a logarithmic market scoring rule; won the main prize among nearly 1,000 participants (Python).

WALKTHROUGH - JUNCTION

Nov 2025

- Built a real-time AI grocery assistant for Snap Spectacles; placed Top 5 overall at Junction and 2nd in the Snap Inc. AR Challenge (Lens Studio, Go, Google Cloud).

TECHNICAL SKILLS AND LEADERSHIP

Programming: Python, TypeScript, JavaScript, Swift, Scala, SQL, R, Go

Technologies: PyTorch, React, Oracle APEX, Neo4j/Cypher, SQLite, Git, GitHub Actions, MCP, Google Cloud

Awards: Hack & Roll Singapore main prize; Junction Top 5 and Snap AR 2nd place; Dean's Scholarship (2024, 2025)

Leadership: Aalto Triathlon Club CFO; Slush Team Lead; Finnish Marines NCO course valedictorian (#1/200)